The Insufficiency of Static Word Vectors

Navigating the Path of Meaning

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The assumption sitting underneath much of modern AI is that a word can be represented by a single vector. And it appears on first inspection that this works. Models like Word2Vec, GloVe,

and fastText gave us elegant geometric spaces where words can be placed in a space and appear to show that meaning itself had been captured. However, when we look deeper we find something maybe missing. ‘Meaning’ may not what it initially seems

When we look close, we find that the problem isn’t that static embeddings are “Simple”. It’s that they are **structurally incapable** of representing meaning. Not suboptimal or outdated, but fundamentally insufficient**.** This isn’t a claim about training data or model size but as we will see it’s a consequence of three independent constraints:

Firstly within static vectors information is destroyed. If we take a word like *“bank” it* has multiple meanings:

- river bank
- financial bank
- banking an airplane

yet a static embedding assigns **one vector** to all of them. That means that different meanings → same representation. Once that happens, the system cannot recover which meaning was intended. The information is gone, not hidden and not compressed in a clever way; it is simply **irreversibly lost.**

Secondly, we know by measurement that meaning is not a singularity and a point — it’s a trajectory. To make this clear we just have to see that language unfolds through time. Each word depends on what came before and what comes next. Meaning isn’t a location in space; it’s a **path through a space**. When “bank” appears in a sentence, it doesn’t sit still — it *moves*:

- toward finance
- toward geography
- toward aviation

depending on context.

So we can see that static embeddings collapse all of those paths into a single point. We can see this as like trying to understand a dance from a single photograph. The motion — which *is* the meaning and the dance — disappears.

Thirdly, it's important that we can show how the chain of meaning is broken. consider that before words are text, they are sound. That sound is made up of continuous acoustic patterns shaped by the vocal system. These patterns are carefully compressed into:

sound → symbols → tokens

Each step preserves just enough structure to reconstruct meaning. However, static embeddings perform a final step:

tokens → fixed vectors

And in doing so, they average across all uses of a word. This doesn't just simplify. It breaks the chain and the connection between meaning and its underlying structure is severed.

Yet, we must ask why do modern language models work because they clearly do. These models, like BERT and GPT, succeed because they don't rely on static representations. They generate **different vectors for different contexts, this is fundamental to the attention mechanism that lies at the core of modern LLM**. In plain words: they restore part of the trajectory. They begin to fix the problem of static vectors by adding in relative motion and that's the magic movement that gives language meaning.

A Different Way to Think About Meaning

Those that have followed my work on modelling language as dynamical system that takes these constraints seriously, will have see the emergence of a new picture where:

- Meaning is not stored in words
- It is reconstructed through **sequences**
- It lives in the **geometry of transitions**, not static positions

This suggests that future systems should not focus on better word vector, but on **better trajectory reconstruction**. This has been the focus on my work of building a trajectory based AI and Language Model.

The Full Argument

This article is just an introduction to the idea that we formally need dynamic vectors to hold representational meaning. The full work develops three formal proofs:

- Information-theoretic impossibility
- Dynamical systems insufficiency (via Takens' theorem)
- Transduction chain violation

and shows how they converge on the same conclusion. If this line of thought resonates, the full paper develops the argument rigorously across information theory, dynamical systems, and signal transduction.

👉 **Read the full paper here:** *Static Vector Insufficiency for Natural Language Meaning*

Final Thought

Perhaps, as your own mind navigates the meaning in this article and even examines the formal proofs. You may see that we previously failed to find the dynamical vector, the motion of of reading, writing and conversation because we asked the wrong question. As meaning is not a static point in space, it is a dynamical living, motion. It was and is always a path. You are walking that path with me now.

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> *Everything that exists is finite; it can only be known by measure*
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